EX:NO:16

PROGRAM:

```
def champagneTower(poured, query_row, query_glass):
    dp = [[0.0] * (k + 1) for k in range(query_row + 1)]
    dp[0][0] = poured

    for r in range(query_row):
        for c in range(len(dp[r])):
            overflow = (dp[r][c] - 1) / 2
            if overflow > 0:
                dp[r + 1][c] += overflow
                dp[r + 1][c + 1] += overflow

    return min(1, dp[query_row][query_glass])

# Test
print(champagneTower(1, 1, 1))  # Output: 0.0
print(champagneTower(2, 1, 1))  # Output: 0.5
```

Output

```
0.0
0.5

=== Code Execution Successful ===
```